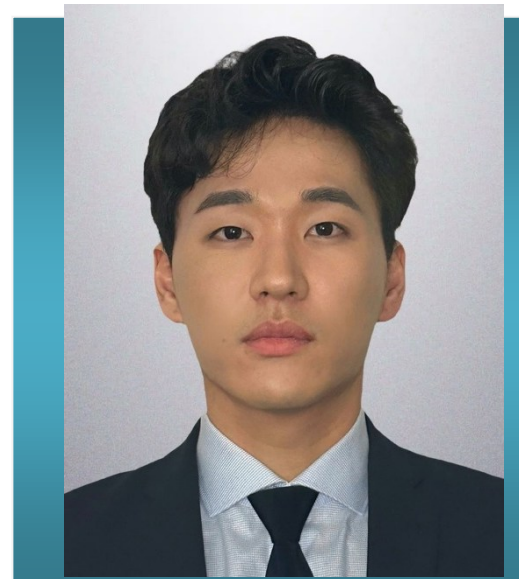


CURRICULUM VITAE IN-HO BAEK

In-Ho Baek 백인호 M.Sc.

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PERSONAL INFORMATION

Date of Birth: 28th April 1995
Place of Birth: Wolfenbuettel, Germany
Marital Status: single

RELEVANT PRACTICAL EXPERIENCES

07/2021 – ongoing

Fraunhofer Heinrich Hertz Institute HHI, Berlin
Scientist, Full-Time

- Research in the field of optical communications with focus on terahertz communications incl. intentions of obtaining a Ph.D. (Dr. Ing.)
 - Advanced DSP development for rate-adaptive THz wireless communications
 - DSP on chip for high-speed coherent 800G ASICs
 - Device & setup characterization
 - Project management

10/2025 – 03/2026

HTW Berlin University of Applied Sciences, Berlin
Lecturer

- Lecturer for the Bachelor's degree program Information and Communication

10/2022 – 03/2024

Berlin University of Applied Sciences and Technology, Berlin
Lecturer

- Lecturer for the Master's degree program Information and Communication

10/2020 – 04/2021

Institute of Photonics and Quantum Electronics (IPQ), KIT, Karlsruhe
Master Thesis

- Data-Aided Carrier Phase Recovery Algorithms for Coherent Optical Communication Systems

04/2020 – 06/2021

EnBW AG
Working Student

- Rheinhafen Power Plant – Risk assessments of electrical systems according to the Ordinance on Industrial Safety and Health specified by EU Directives and related harmonized standards
- Calculation, reconstruction and infrastructure adjustments of the power plant HV site grid
- Identification, analysis and designation of explosion risk areas

04/2019 – 03/2020	Siemens AG <u>Working Student</u> ➤ Digital Industries – Factory Automation – Operational Services ➤ Construction and maintenance of kiosk systems based on Linux OS
06/2017 – 11/2017	Institute of Micro- and Nanoelectronic Systems, KIT, Karlsruhe <u>Bachelor Thesis</u> ➤ Investigation of the timing jitter of superconducting single photon detectors with varying meander shapes
04/2016 – 09/2016	Institute of Micro- und Nanoelectronic Systems, KIT, Karlsruhe <u>Tutor/teaching assistant</u> ➤ Module: Electronic Circuits ➤ Teaching of hands-on knowledge to students ➤ Correction and grading of exercise sheets

EDUCATIONAL BACKGROUND

07/2021 – 07/2026	Technische Universität Berlin (TUB), Berlin <u>PhD: Electrical Engineering/ Information Technology</u> ➤ Current Working Title: Investigating Adaptive Baseband Techniques for Converged Optical / THz Wireless Links ➤ Officially registered at TUB on 24th July 2024. ➤ Desired degree: Dr.-Ing.
04/2018 – 04/2021	Karlsruhe Institute of Technology (KIT), Karlsruhe <u>Graduate/Master Studies: Electrical Engineering/ Information Technology</u> ➤ Master Thesis: <i>Data-Aided Carrier Phase Recovery Algorithms for Coherent Optical Communication Systems</i> ➤ Focal points: Communication Engineering / Optical Communications
08/2018 – 02/2019	Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea <u>Exchange program Masterstudies: Electrical Engineering & Philosophy</u> ➤ Course 1: Radar Systems Engineering ➤ Course 2: MIMO Wireless Communications ➤ Course 3: Philosophy of Science
10/2013 – 03/2018	Karlsruhe Institute of Technology (KIT), Karlsruhe <u>Undergraduate/Bachelors Studies: Electrical/Information Engineering</u> ➤ Bachelor Thesis: <i>Investigation of the timing jitter of superconducting single photon detectors with varying meander shapes</i>
06/2013	Marienschule, Hildesheim <u>Abitur (German High School Diploma)</u>

PUBLICATIONS

First Author:

In-Ho Baek, Julian Honecker, Marie Bittiehn, Robert Elschner, Colja Schubert, and Ronald Freund, "*Experimental Demonstration of Neural-Network Based Transceiver Nonlinearity Mitigation for Probabilistically Shaped Photonic THz Wireless Transmission*," in Proc. Optical Fiber Communication Conference (OFC), 2026. **SUBMITTED & IN REVIEW**

In-Ho Baek, Oliver Stiewe, Jonas Gläsel, Simon Nellen, Garrit Schwanke, Trung T. Tran, Robert Elschner, Colja Schubert, Robert B. Kohlhaas, Patrick Runge, Martin Schell, Ronald Freund, "Experimental Characterization of Photodiode Transmitters for Probabilistically Shaped High-Speed THz Wireless Transmission," *Journal of Lightwave Technology*, 2025.

IH Baek, O Stiewe, J Gläsel, MF Ulukan, A Schindler, F Ganzer, TT Tran, "Experimental Characterization of a WR3-Coupled Photodiode Transmitter for High-Speed Terahertz Wireless Communication," 2024 International Topical Meeting on Microwave Photonics, 2024.

IH Baek, O Stiewe, S Nellen, G Schwanke, R Elschner, C Schubert, "32 GBd, 109 Gbit/s probabilistically shaped THz wireless transmission using PIN-PD based photonic upconversion," IET Conference Proceedings CP839 2023 (34), pp. 1302-1305, 2023.

IH Baek, F Bart, R Elschner, F Meier, D Hellmann, A Maassen, C Schubert, "Time adaptive probabilistic shaping for combined optical/thz links," Photonic Networks; 23th ITG-Symposium, pp. 1-8, 2022.

Co-Auth:

Robert Elschner, Frederik Bart, Fred Meier, Oliver Stiewe, **In-Ho Baek**, Simon Schütze, Andreas Maaßen, Kallyan Das, Robert B. Kohlhaas, Simon Nellen, Milan Deumer, Thomas Merkle, Axel Tessmann, Markus Rösch, Colja Schubert, Ronald Freund, and Martin Schell, "*Analysis of Weather-Related Link Availabilities for Fiber-Integrated THz-Wireless Transmission Systems*," *Journal of Lightwave Technology*, 2025

C Schubert, R Elschner, O Stiewe, **IH Baek**, S Schütze, A Maaßen, K Das, "High-Capacity THz Wireless Transmission Supporting Future 6G Optical Networks," Optical Fiber Communication Conference, M3D. 1, 2025.

C Schmidt-Langhorst, J Pfeiffer, R Elschner, R Emmerich, F Chowanek, **IH Baek**, Robert FH Fischer, C Schubert, "Experimental Investigation of Information Bit Scrambling for Physical-Layer Security in Coherent Fiber-Optic Systems," Photonic Networks; 23th ITG-Symposium, pp. 1-6, 2022.

LANGUAGE SKILLS

German	native
Korean	upper intermediate
English	fluent in spoken and written

OTHER QUALIFICATIONS

Social/Sport

- Former volunteer at Lebenshilfe Karlsruhe e.V. for disabled people
- Former volunteer at SiMaBo Cabo Verde (animal welfare)
- Former honorary tutor for children from difficult circumstances

Scholarships

Baden-Wuerttemberg Scholarship 2018